Episode 8: The Data Room

Programming with Python · Tutorial 08



The investor wasn't impressed by a slide. She was impressed by *data*. So she slides a USB stick across the table: **every order, two full weeks.** "Impress me." That stick is now a real file, `orders_messy.csv`. Tobi exported it from the ordering system and nobody has looked at it since. It has too many rows to eyeball and too many for a hand-written loop, so this week you use the tool built for exactly this: **pandas**. You'll load the file into one *DataFrame*, filter it with boolean masks, add a derived column on a safe copy, and answer per-zone questions with `groupby`. Tobi, meanwhile, has discovered AI. Your real job is to **supervise** it and catch the confident nonsense.

Run the lab on your laptop

From this episode the lab is a Python script that runs on your own machine, in the `python-labs` folder you created in Session VII. Missed that session? The Tutorial 07 page says how to catch up.

1. Download both files and move them into your `python-labs` folder:

   Lab script Orders data

2. In Zed, open the `python-labs` folder, then the terminal (`Ctrl+`).
3. Run `uv run python lab_08_dataroom.py`.

> **No Zed on your laptop, or only an iPad?**
> Use a **GitHub codespace**: a Linux computer that runs in a browser tab.
>
> 1. Sign in at github.com/codespaces and start the **Blank** template. If you set one up in Session VII, open it again and go to step 3.
> 2. In its terminal, follow the uv guide (the macOS or Linux lines) down to `uv add pandas matplotlib`.
> 3. Go into the folder with `cd python-labs` and download both files:
>
>    ```
>    curl -O https://python.tobiasvlcek.com/labs/lab_08_dataroom.py
>    curl -O https://python.tobiasvlcek.com/labs/orders_messy.csv
>    ```
>
> 4. Run `uv run python lab_08_dataroom.py`.
>
> A codespace has no Zed, so keep the chatbot open in a second tab for AI help.

The first run prints three tables and then stops with `AssertionError: Task 1: not attempted yet.` That is the lab working: the script checks every task and stops at the first one whose answer is missing or has the wrong form. Solve it, save, and run again.

If the run ends in `ModuleNotFoundError: No module named 'pandas'` instead, the terminal is not inside `python-labs`, or the packages are missing. In that folder, run `uv add pandas matplotlib` once.

> ! **Look at your numbers**
> Each task gives you a symptom (“Hafen is missing orders”), and you find the cause. The checks only test the form of your answers, so a wrong number can pass all of them. And **one problem in the export is not mentioned anywhere**: no task names it and no check catches it.

When every check passes, the script prints a block that starts with `SHOW THIS IN CLASS`. Show it to your lecturer, who compares your numbers with the right ones. That is feedback, not a condition for leaving: if a number differs, you hear which one, and whether you fix it is up to you. Not finished when the session ends? Finish at home. You do not need to show the block later.